

POWERFACTORY

PowerFactory 2024

Technical Reference

AC/DC Connector

ElmConnectacdc

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

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January 9, 2024
PowerFactory 2024
Revision 1

Contents

1 General Description **1**

1 General Description

The *AC/DC Connector* element in *PowerFactory* is used for connecting an AC node to a DC node. It is a purely EMT single-phase model.

It is possible to define a resistance and an inductance, in which case the AC/DC connector is equivalent to a series reactor connected between an AC and a DC node.

The equations describing the model are:

$$u_{AC} - u_{DC} = R \cdot i_{AC} + L \cdot \frac{di_{AC}}{dt} \quad (1)$$

$$i_{AC} = -i_{DC} \quad (2)$$

where the currents i_{AC} and i_{DC} are both defined positive if entering the AC or DC node.